HDI INSIGHT AI

TECHNICAL INSTALLATION & DEPLOYMENT GUIDE

Target Environment: Local / Server	**Software Version:** v1.0.0
Dependencies: Python 3.10+	**Document Classification:** Public

1. System Overview & Requirements

This guide outlines the system specifications and step-by-step setup instructions to install, initialize, and deploy the **HDI Insight AI (Human Development Analytics Platform)** codebase. The application is built using a Flask backend, SQLite database, and scikit-learn machine learning binaries, making it highly portable and standard-compliant.

Hardware Requirements:

- **CPU:** Dual-core 2.0 GHz or higher (x86/x64 or Apple Silicon)
- **RAM:** 4 GB minimum (8 GB recommended for model training)
- **Disk Space:** 100 MB free space (excluding training datasets)

Software Dependencies & Prerequisites:

- **Python Version:** Python 3.10 or 3.11 is recommended. Validate with python --version.
- **Database Engine:** SQLite3 (Standard library, no external server required).
- **Web Browser:** Google Chrome, Mozilla Firefox, or Microsoft Edge (recent versions).

2. Step-by-Step Installation

Step 1: Obtain the Codebase

Clone the repository from GitHub or extract the project source files to your local system:

```
git clone https://github.com/pandu0426/HDI-Insights.git
cd HDI-Insights/05_Project_Development
```

Step 2: Create a Python Virtual Environment

It is strongly recommended to isolate the project dependencies within a local virtual environment. Run the following command inside the project development folder:

```
python -m venv venv
```

Step 3: Activate the Virtual Environment

Select and run the appropriate activation command for your Operating System's shell environment:

Windows (PowerShell):

```
venv\Scripts\Activate.ps1
```

Windows (Command Prompt):

```
venv\Scripts\activate.bat
```

macOS / Linux (Bash/Zsh):

```
source venv/bin/activate
```

Step 4: Install Required Packages

Install the compiled web application dependencies, including Flask, Pandas, NumPy, Scikit-Learn, Joblib, and ReportLab:

```
pip install -r requirements.txt
```

Step 5: Train and Serialize ML Models

Run the machine learning preprocessing, training, and benchmarking pipeline. This cleans the raw UNDP dataset, performs log transforms, trains the classifier and regressor companion models, outputs comparative JSON stats, and serializes the .joblib models:

```
python train_model.py
```

Step 6: Start the Flask Application Server

Initialize the SQLite history database and launch the WSGI development web server:

```
python app.py
```

3. Verification and Testing

Once the Flask server is running, you should verify its operation by executing the following procedures:

1. Open your web browser and navigate to: **http://localhost:5000/**.
2. Verify that the homepage dashboard renders without errors and displaying the historical chart logs.
3. Open another terminal panel, activate your environment, and execute the automated unit test suite:

```
python -m unittest test_app.py
```

4. Troubleshooting Common Issues

Error: Missing Dependencies / Libraries

If you encounter a 'ModuleNotFoundError', ensure your virtual environment is actively running in the current terminal window. Run 'pip list' to confirm the installed dependencies.

Error: Address already in use (Port 5000)

If Flask crashes with a bind error (port already in use), another server is running on port 5000. You can change the port in the app run settings inside app.py, or run: python app.py --port=5001.